



SATHWIKAJ BANDARU

AI Systems Master's Student
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LinkedIn | GitHub | Portfolio

PROFILE

Master's student in Artificial Intelligence Systems at EPITA with hands-on experience building machine learning models, NLP applications, Generative AI systems, and multi-agent workflows using Python and modern AI frameworks. Built production-grade Retrieval-Augmented Generation (RAG) pipelines, MLOps systems with Airflow orchestration, and deep learning models for healthcare diagnostics. Interested in AI engineering, applied machine learning, and intelligent systems for real-world analytical applications.

TECHNICAL SKILLS

Programming	Python, SQL, JavaScript
ML & Analytics	Scikit-learn, SHAP, Predictive Analytics, Statistical Analysis, Data Mining, Model Evaluation
Deep Learning	PyTorch, TensorFlow, CNNs, Transformers, Grad-CAM
NLP & GenAI	NLP, LLMs, RAG, FAISS, Generative AI, Agentic AI, Multi-Agent Systems, LLM-as-Judge Evaluation, Conversational AI
Data & Storage	Pandas, NumPy, PostgreSQL, ETL Pipelines, Great Expectations
Backend & APIs	FastAPI, Streamlit, REST APIs
MLOps & Tools	Apache Airflow, Docker, Docker Compose, Git, CI/CD

PROJECTS

- LLM-Based RAG + Multi-Agent AI Research Assistant** | *Python, LLMs, FAISS, NLP* | [GitHub](#) 2025
 - Engineered a Business Law RAG Assistant using Python, LLMs, and FAISS vector indexing for natural language Q&A over legal documents.
 - Implemented an LLM-as-Judge evaluation suite with chain-of-thought reasoning to verify answer faithfulness, achieving **100% accuracy (19/19)** across Factual, Cross-reference, Out-of-scope, and Ambiguous question categories.
 - Built automated API cost tracking with per-script logging to monitor production usage and optimize prompt efficiency across the retrieval pipeline.
 - Designed modular multi-agent workflows for retrieval, generation, and evaluation to improve contextual response quality and reliability.
- Real-Time Bike Demand Forecasting Platform** | *Airflow, PostgreSQL, FastAPI, Docker, Great Expectations* | [GitHub](#) 2024
 - Architected an end-to-end MLOps pipeline with **4 Apache Airflow DAGs** (ingestion, training, prediction, data generation) running at **2-minute prediction intervals**, deployed via Docker Compose with FastAPI, Streamlit, and PostgreSQL services.
 - Implemented automated data validation using **7 Great Expectations rules** with a 3-tier criticality classification (HIGH >50%, MEDIUM 10–50%, LOW <10% invalid rows) to route incoming files into good/bad data partitions.
 - Designed idempotent pipelines via PostgreSQL state tracking (`processed_files` table) and XCom-based file management to prevent duplicate processing across DAG runs.
- Clinical Text Intelligence System** | *Python, NLP, FastAPI, Streamlit* | [GitHub](#) 2024
 - Built an NLP-driven application for healthcare text analysis, entity extraction, automated summarization, and clinical data interpretation.
 - Developed FastAPI backend services and Streamlit interfaces to streamline conversational AI interactions for clinical workflows.
 - Achieved an F1-score of **0.92** for named entity recognition using NLP-based information extraction and model optimization.
- Disease Risk Prediction System** | *Machine Learning, SHAP, Predictive Analytics* | [GitHub](#) 2024
 - Developed Python-based ML models for disease risk prediction using clinical datasets to support data-driven healthcare decision-making.
 - Applied SHAP interpretability, feature engineering, and statistical analysis to improve transparency of AI predictions.
 - Achieved an AUC-ROC score of **0.89** while optimizing classification performance across multiple disease scenarios.
- Medical Image Diagnosis System** | *Deep Learning, CNN, Grad-CAM* | [GitHub](#) 2024
 - Designed a CNN-based medical image diagnosis system with Grad-CAM interpretability, achieving **97% accuracy** and **95.4% recall**.
 - Improved explainability of diagnostic predictions using Grad-CAM visualization and deep learning optimization techniques.

EDUCATION

- EPITA** 2025 – Present
Master's – Artificial Intelligence Systems
Relevant coursework: Neural Networks & Deep Learning, Natural Language Processing, Machine Learning (Statistical, Bayesian & Unsupervised), Computer Vision, Reinforcement Learning, Time-Series Analysis, Recommender Systems, Big Data Infrastructure & Cloud Computing, Data Science in Production
- Kalasalingam University of Technology** 2021 – 2025
Bachelor of Technology – Artificial Intelligence & Machine Learning
- Politechnika Krakowska** 2024
International Exchange Semester – Artificial Intelligence & Machine Learning

PUBLICATIONS

Object Identification for Visually Impaired Using IoT with Image Processing — Published research on AI-assisted computer vision and IoT systems for accessibility-focused applications using image processing and automation techniques.

SOFT SKILLS & LANGUAGES

Soft Skills	Analytical Problem Solving, Technical Communication, Cross-functional Collaboration, Adaptability, Technical Documentation
Languages	English (Fluent), French (Beginner), Telugu (Native), Hindi (Fluent)